

Lecture 12 Notes: Bonds - Types, Features, and Market Dynamics

Corporate Finance Chapter 7

I. Assignments and Review

- The assignment on bond valuation questions should be submitted on Google Classroom by tomorrow.
- A submission area will be created under “Classwork.”

II. Major Categories of Bonds

Bonds are categorized based on the issuer. Each type carries distinct risk characteristics.

1. Treasury Bonds

- Issued by the federal government.
- **Key Feature:** Considered to have **no default risk** because the government can tax or print money.
- Short-term Treasuries have no maturity risk, but long-term bonds are still subject to interest rate risk.

2. Corporate Bonds

- Issued by companies.
- **Key Feature:** Carry **default risk** (credit risk) because companies can face financial distress or cash flow problems.
- As default risk increases, investors demand a higher expected return. This is reflected in the **default risk premium**.
- Lower-rated (junk) bonds offer higher yields to compensate for this risk.

3. Municipal Bonds

- Issued by state and local governments (primarily in the U.S.).
- **Key Feature:** Coupon payments are **exempt from federal taxes**.
- While rare, state governments can default (e.g., due to budgetary deficits).

4. Foreign Bonds

- Issued by foreign governments or corporations.
- **Key Features:** Involve **sovereign risk** and, if denominated in a foreign currency, **exchange rate risk**.

III. Yield to Maturity (YTM) Revisited

Definition

The **Yield to Maturity (YTM)** is the **expected annual return** an investor will earn if the bond is held until maturity. It is often called the “**promised yield**” because it is the yield promised by the bond contract.

Relationship Between Coupon Rate, YTM, and Bond Price

- When a bond is issued, the coupon rate is set close to the required return (YTM), so the bond sells at **par** (\$1,000).
- If market interest rates rise, the YTM increases. The bond's fixed coupon becomes less attractive, and the price falls below par (**discount**).
- If market interest rates fall, the YTM decreases. The bond's fixed coupon becomes more attractive, and the price rises above par (**premium**).

Key Relationship

Bond Prices and Interest Rates (YTM) move in **opposite** directions.

- **Inverse Relationship:** As YTM $\uparrow$, Bond Price $\downarrow$.
- **Inverse Relationship:** As YTM $\downarrow$, Bond Price $\uparrow$.

IV. Special Bond Features

Callable Bonds

- A **call provision** gives the **issuer** the right to redeem (repay) the bond before its maturity date.
- **Why call?** If interest rates fall, the issuer can call the old bonds (paying a high coupon) and refinance by issuing new bonds at a lower rate.
- **Cost to Investor:** The investor loses the future high coupon payments and faces **reinvestment risk** (having to reinvest at lower rates).
- **Compensation:** Issuers typically pay a **call premium** (e.g., \$1,100 instead of \$1,000) to compensate for this risk.

- Callable bonds generally offer a **higher yield** than non-callable bonds to compensate investors for the call risk.

Sinking Fund

- A provision requiring the issuer to **retire a portion of the bond issue each year** over the bond's life, rather than repaying the entire principal at maturity.
- **Benefit to Investor:** Reduces default risk because the principal is repaid gradually.
- **Risk to Investor:** If interest rates have fallen, the investor receives principal earlier and must reinvest at lower rates (**reinvestment risk**).

V. Economic Context: Default Risk in Practice

The discussion moved to the practical implications of default risk, using Pakistan as a case study.

Class Discussion: Government Default and Currency Risk

Student Question: If a government can print money to avoid default, doesn't that cause devaluation? Doesn't that make the investor worse off?

Instructor: Yes, that is correct. While there is no nominal default risk, there is still **interest rate risk** and **inflation risk** (purchasing power risk). You are worse off in real terms, but you would be "more" worse off in a total default scenario.

Context on Pakistan's Debt:

- Debt repayments are a major part of the national budget.
- A \$1 billion bond is maturing on April 26.
- **Why no default?** During Eid season, remittances from overseas Pakistanis are expected to reach \$4.5–\$5 billion. The government can purchase dollars from the market using rupees to meet its obligations.
- The immediate default risk has decreased due to an improved **current account deficit**.

VI. Summary of Key Concepts

1. **Bond Types:** Treasury (no default risk), Corporate (default risk), Municipal (tax-exempt), Foreign (sovereign + currency risk).
2. **YTM:** The promised return if held to maturity.
3. **Price-YTM Relationship:** Inverse.
4. **Call Provision:** Issuer's right to repay early; leads to higher yields.

5. **Sinking Fund:** Gradual repayment; reduces default risk but increases reinvestment risk.